

Interactions among Unfamiliar Spiny Mouse (*Acomys cahirinus*) Weanlings

RICHARD H. PORTER, VICKI J. TEPPER, ALAN A. BAUMEISTER,
JENNIFER M. CERNOCH, AND JOHN A. MATOCHIK¹

*The John F. Kennedy Research Center, George Peabody College, Vanderbilt University,
Nashville, Tennessee 37203*

Social interactions among unfamiliar *Acomys cahirinus* pups are observed following removal from the home cage and parents. Huddling by 14- to 22-day-old pups declines sharply as the age of the pups increases. This reduced responsiveness to unfamiliar agemates coincides with a downward trend in suckling rate from Days 16 through 22, indicating that with the approach of final weaning animals become more discriminating in choice of social partners. Compared to singly reared pups, those reared with 2 to 3 littermates subsequently display more frequent huddling with unfamiliar agemates and are more successful in a food competition test. Experience with littermates appears to result in increased social competence after weaning.

Weaning is characterized as a period of major transitions in the life of mammals (e.g., Galef, 1981; Swanson & Campbell, 1980). As the weaning process proceeds, mammalian young undergo a series of physiological and behavioral changes and also experience drastic alterations in their social relationships. Mothers and offspring may come into conflict over the age at which nursing should be discontinued (Trivers, 1974), weanlings of many species tend to disperse from their parents and littermates (and original home area), and contact with conspecifics other than members of the immediate family may become increasingly frequent.

Despite the importance of appropriate responsiveness to conspecifics for the ultimate reproductive success of weanlings, few studies have focused specifically upon social interactions involving unfamiliar animals during this stage of life. Previous research in our laboratory has shown that spiny mouse (*Acomys cahirinus*) weanlings respond differentially to littermates as contrasted with unfamiliar agemates when both classes of conspecifics are simultaneously present (Porter, Wyrick, & Pankey, 1978; Porter & Wyrick, 1979). Social interactions among unfamiliar spiny

¹ The research project reported in this paper was supported in part by NICHD Grant 00974 and training Grant HD 07226.

mouse weanlings were investigated further in the present series of studies that were specifically concerned with huddling as a function of age of pups, and the interactions between pups that had been reared with, vs without, littermates.

I. AGE-DEPENDENT HUDDLING BY UNFAMILIAR PUPS

When group housed in the laboratory, various species of rodents have been observed to spend considerable periods of time in physical contact with conspecifics (i.e., clumped or huddled) (e.g., Calhoun, 1962; Parker, Smith, Erickson, & Cromer, 1980). Norway rats, for example, maintain frequent bodily contact beginning shortly after birth and continue to huddle and sleep together throughout the juvenile period and adulthood (Alberts & Brunjes, 1978). Dyadic huddling has recently been used as a measure of social preferences in studies of sibling recognition in spiny mice (Porter & Wyrick, 1979). Whereas 9- to 11-day-old pups huddle indiscriminately in groups containing familiar siblings as well as unfamiliar nonsiblings, preferential pairing by siblings occurs by 14 to 15 days of age. The initial experiment in the present report was conducted to assess the development of responsiveness among unfamiliar pups in the *absence* of familiar littermates.

Methods

Subjects were randomly selected from litters of 2 to 4 pups born in the laboratory colony of *A. cahirinus* where mated pairs are housed together continuously in individual breeding cages. All offspring remained with their parents and littermates until the beginning of testing. At that time, 2 unrelated pups born on the same day were removed from their respective home cages and placed in an observation terrarium measuring 60 × 30 cm. The floor of the terrarium was covered with a thick layer of bedding material and Purina mouse chow and pieces of fresh apple were available ad libitum.

Each pair of weanlings was observed over a 2-day period with the first observation session beginning 60 min after the animals were placed together (to allow time for them to explore and become somewhat familiar with their new environment). Data were collected during 6 observation sessions on each of the 2 days. At the beginning of each 5-min observation session, and at the end of each 60-sec interval therein, the two pups in a given cage were scored as being either separate or in physical contact with one another (i.e., touching of head, limbs or trunks, excluding touching of whiskers or tails). Therefore, for each 5-min session there were a total of 6 observation points. The six 5-min sessions were distributed throughout the day with a minimum interval of 30 min between successive sessions. Over the 2 days of observation there was a total of 72 sample points for each pair of weanlings.

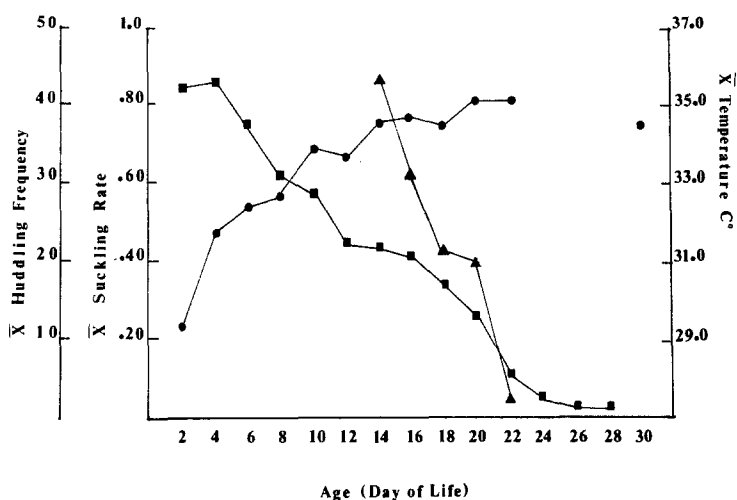


FIG. 1. Ontogenetic changes in frequency of huddling (▲) by unfamiliar animals, suckling rate (■), and body temperature (●).

Forty pairs of pups were tested in the above manner, with 8 pairs assigned to each of 5 age conditions—i.e., testing beginning on Day 14, 16, 18, 20, or 22 postpartum. Day 14 was selected as the youngest age for testing since this is the time when pups first begin to display preferential huddling with familiar as compared with unfamiliar agemates. Within each age condition the sex composition of the 8 pairs was systematically balanced so that 2 to 3 pairs were comprised of each of the three possible combinations (i.e., 2 males, 2 females, or 1 male/1 female per pair).

Results

The mean frequencies of touching across the 5 age conditions are presented in Table 1 and included in graphic form in Fig. 1. As evident from Table 1, with increasing age there was a steady decline in the frequency of touching by animals that were unfamiliar with one another prior to the test sessions. Statistical analysis (ANOVA) revealed an

TABLE 1
Mean Frequencies of Touching (Huddling) by Unfamiliar Pups of 5 Different Ages

	Age (day postpartum at start of 2-day test session)				
	14	16	18	20	22
Mean huddling frequency	43.9	31.8	21.7	20.8	4.4
SD	13.1	30.7	29.6	30.1	12.4

overall significant difference between the mean touching frequencies ($F(4, 35) = 2.77, p < .05$).

TEMPERATURE REGULATION

Alberts (1978) has reported that huddling attenuates temperature loss in 5- to 15-day-old rat pups tested at room temperature. By 20 days of age, however, the body temperature of isolated pups did not differ from that of huddled pups. Subsequent work indicates that huddling is mediated by thermal cues at 10 days of age and younger, while olfactory stimuli from conspecifics become more salient for huddling by Day 15 (Alberts & Brunjes, 1978). It was therefore concluded that huddling changes from a physiological to a social response as pups become more effective in regulating their own body temperature.

By taking daily temperature readings an attempt was made to determine whether the ontogenetic pattern of huddling by unfamiliar *A. cahirinus* pups (reported above) coincides with their ability to regulate their body temperature.

Methods

Temperatures of an initial group of 14 30-day-old pups were recorded to establish a baseline against which to compare the temperatures of younger animals. At 30 days of age, the precocious *A. cahirinus* pups are entirely independent of maternal care—i.e., pups can be weaned successfully at 8 days of age in the laboratory and the eating of solid food has been observed as early as Day 4 (Porter, Cavallaro, & Moore, 1980).

Beginning on Day 2 postpartum daily temperature readings were recorded for a further 18 animals. Such recording was continued Monday through Friday each week until 3 days after the mean temperature of the pups first came within 0.25 C° of the 30-day comparison group.

All animals were removed from their home cages and placed into individual isolation cages at room temperature (21.5–23.0°C) for 60 min prior to recording their temperatures. Immediately following the test session, pups were returned to their respective home cages. Body temperatures were measured with a small surface probe thermistor (Cole-Parmer Model 8443-20) coupled to a Yellow Springs telethermometer (Model 43 TA). Temperature readings were taken by placing the probe against the inner surface of a pup's ear while loosely holding the animal and probe lead in gloved hands. The decision to record surface rather than rectal temperature was based upon evidence indicating that the latter method is more stressful and yields less reliable measures (Hennessy, Li, Lowe, & Levine, 1980; Hutchings, 1967).

Results

Mean body temperatures from Day 2 onward are presented over 2-day blocks in Fig. 1. A fairly constant increase in temperature was observed until Day 14 (when the 30-day baseline temperature was first reached), with the readings remaining stable from Days 14–18—i.e., within 0.4 C° of the 30-day temperature.

SUCKLING FREQUENCY

While suckling, littermates necessarily come into physical contact with one another. As the rate of suckling declines with the approach of weaning there will be a concomitant decrease in such forced bodily contact with littermates as well as with the mother. Since contact with conspecifics is no longer required when obtaining food (nor for thermoregulation), it is hypothesized that weanlings could afford to become increasingly discriminating in regard to choice of social partners.

To determine whether suckling frequency is correlated with general social responsiveness, the rate of suckling throughout the nursing period was monitored for comparison with the above-presented ontogenetic trends in huddling by unfamiliar pups.

Methods

Suckling was observed in 16 family groups, with eight of these groups consisting of mother, father, and two *male* offspring, while the remaining eight groups were each comprised of two parents plus two *female* offspring. On the day of parturition, animals in each group were placed in a glass terrarium (25 × 51 × 25 cm) containing Purina Mouse Chow and apple chunks ad libitum. Data collection began on the second day postpartum and continued from Monday through Friday during the first 30 days after birth.

On every observation day the number of pups suckling (0–2) at each of 10 scan sample observations was recorded for each group. These instantaneous scan samples were conducted at random times from 0745–1800 (during the light phase of the 13/11 hr light/dark cycle) with an interval of at least 30 min between successive samples.

Results

Daily suckling rates were calculated for each pair of pups by dividing their total observed frequency of suckling over the maximum possible score of 20 (i.e., 2 pups × 10 observations/day). Theoretically, suckling rates calculated in this manner could range from 0 through 1.00, with the latter score indicating that at each observation point on that particular day both pups were observed suckling. Since the mean suckling rates for the male and female litters did not differ significantly, the data from

all 16 groups of animals were combined and presented over 2-day blocks in Fig. 1.

Discussion

When combined with the results of previous related studies, the huddling data from the first experiment suggest 3 stages in the development of social responsiveness in *A. cahirinus* pups. Prior to 14 days of age (i.e., through at least Day 11), pups huddle indiscriminantly with familiar and unfamiliar agemates (Porter & Wyrick, 1979). While preferential huddling with familiar conspecifics begins at Day 14 (i.e., littermates are selected over unfamiliar pups when both classes of conspecifics are present), the above experiment indicates that 14 to 20-day-old pups will huddle with unfamiliar agemates if no other options are available. By 22 days of age, however, positive responding occurs almost solely to familiar animals, with unfamiliar agemates avoided even if littermates (or other familiar agemates) are absent. Pups therefore become increasingly discriminating in regard to choice of conspecifics with whom they will huddle.

Although there appears to be no simple temporal relationship between the development of independent temperature regulation and the decline of huddling by unfamiliar pups (see Fig. 1), it is interesting that Day 14 is the age at which body temperature first reaches the adult level in isolated animals and pups begin to respond preferentially to littermates. As is the case for rats, early huddling by *A. cahirinus* pups (through Day 11) may be a means of temperature regulation, with this response becoming more truly social (and discriminative) as pups gain increasing control over their bodily temperature (Day 14 and beyond) (Alberts & Brunjes, 1978).

As seen in Fig. 1, the decline in huddling among unfamiliar pups closely follows the downward trend in suckling rate from Days 16–22 of life. Thus, as suckling declines so does positive responsiveness to unfamiliar agemates. This increased avoidance of unfamiliar conspecifics with the approach of final weaning may play a role in the social dynamics and system of space utilization of *A. cahirinus*. For instance, avoidance of conspecific agemates may be related to a territorial system of resource apportionment. Such a conclusion is no more than speculation at this time since the necessary field studies of the behavioral ecology of *A. cahirinus* have yet to be conducted.

II. INTERACTIONS AMONG WEANLINGS REARED WITH VS WITHOUT LITTERMATES

Behavioral and physiological differences in pups reared in large vs small litters have been reported for rats (Grotta, 1973; Fleischer & Turkewitz, 1979; Seitz, 1954) and *Mus musculus* (LaBarba & White, 1971;

Preistnall, 1972). There have been few studies, however, that have compared rodent pups raised with littermates against those reared singly. The limited data for both gerbils (Elwood & Broom, 1978) and rats (Leigh & Hofer, 1973) suggest that pups reared in litters of one may develop less rapidly than those in medium sized litters, and may also have a higher mortality rate.

The following series of studies focuses upon social interactions among unfamiliar *A. cahirinus* weanlings raised without littermates vs those from litters of 3 to 4 pups. These litter sizes are within the range normally found in laboratory-bred animals (in the authors' laboratory, litter size has ranged from 1 to 6 pups over approximately 4000 litters, with a median of 2 to 3 pups).

HUDDLING BY UNFAMILIAR WEANLINGS

The influence of experience with littermates on the development of social responsiveness was investigated by comparing huddling in pairs of unfamiliar animals that had been reared singly or with 2 to 3 littermates.

Methods

Huddling within 16 dyads comprised of unfamiliar pups that had been raised *without* littermates was observed as described above. Eight of these dyads were tested on Days 14 and 15 of life of the age-matched pups, while the remaining 8 pairs were observed on Days 16 and 17. The behavior of these 16 pairs of singly reared pups was compared to that of the 16 pairs of weanlings described in the first experiment that had been reared with 2 to 3 littermates each and tested at these same ages (i.e., 8 dyads observed on Days 14 and 15, and another 8 dyads observed on Days 16 and 17). Data were collected concurrently on animals in these two conditions (singly and littermate-reared pups) and sex composition of dyads was equated across the two conditions within each of these two age groups. Further procedural details are otherwise identical to those presented above for the Experiment 1 huddling observations.

Results

Mean huddling frequencies for the littermate and singly reared pairs are presented in Table 2 for each of the two age levels. ANOVAs revealed significant main effects for both rearing condition and age of pups. The pups raised with littermates huddled more frequently than the singly reared weanlings ($F(1, 28) = 17.6, p < .0005$), and the 14 and 15-day pups huddled more than the older (16 and 17 day) pups ($F(1, 28) = 4.3, p < .05$). No statistically reliable age $\times$ rearing condition interaction was obtained.

TABLE 2
Mean Frequencies of Huddling (over a 2-day Observation Period) for Pairs of
Unfamiliar Pups That Had Been Reared Singly or with Littermates

Rearing condition	Age of pups at testing		Total
	14-15 Days	16-17 Days	
Littermate reared	43.9 (13.1) ^a	31.8 (30.7)	37.8 (23.6)
Singly reared	17.0 (21.1)	0.3 (0.7)	8.6 (16.8)
Total	30.4 (21.9)	16.0 (26.5)	

^a Standard deviation in parentheses.

* $p < .05$.

** $p < .0005$.

FEEDING COMPETITION

In a pioneering study of litter size effects, Seitz (1954) reported that adult rats that had been raised in litters of 12 pups were more successful than animals from litters of 6 in a food competition task. Likewise, as a function of early exposure to, and interactions with littermates, weanlings from multiple-pup litters might be expected to have an advantage over pups reared without littermates when confronted with an unfamiliar conspecific agemate. In particular, pups reared with littermates might be less intimidated by the presence of an unfamiliar conspecific and more efficient in interpreting social signals from the latter. This prediction was tested by pairing *A. cahirinus* pups that had been reared with vs without littermates in a food competition test.

Methods

Twenty litters of *A. cahirinus* pups were removed from their parents at 29-31 days of age and placed in individual cages containing ad libitum water but *no* food. Ten of these litters were comprised of only 1 pup each, while the remaining 10 litters each contained 2-4 pups. Twenty-four hours later, the food-deprived weanlings were tested in pairs made up of one animal that had been raised with littermates and one singly reared weanling. Only one pup from any given litter was included in this experiment ($N = 10$). Sex combinations of the 10 pairs of unfamiliar weanlings were randomly determined.

Feeding tests were conducted in a 25 × 50 × 30-cm-high observation terrarium. Food (Purina Mouse Chow) was only available from a cup measuring 8-cm diameter × 4.5-cm high. A 3.3-cm diameter opening centered in the top of a tightly fitting lid provided access to food within

the cup. A perforated metal inner lining positioned between the food pellets and the inside of the lid prevented pellets from being removed from the cup. Thus, in order to feed, animals were required to perch on top of the cup and gnaw at the food pellets through the perforated lining. The size of the cup was such that two animals feeding simultaneously would be forced into physical contact with one another.

At the beginning of a test session, the two animals within a test pair were placed together in an observation terrarium containing a single food cup and observed for 20 min. The following measures were recorded for each animal in the pair: *latency* until climbing onto the food cup, and *feeding duration*. Since it was difficult to ascertain when an animal was actually ingesting food, *feeding duration* was defined as the amount of time spent perched on top of the food cup.

Results

Means for feeding duration and latency until climbing onto the food cup are presented for weanlings from the two rearing conditions in Table 3. Wilcoxon matched-pairs signed-ranks tests indicate that the animals reared in litters of 2 to 4 spent significantly longer periods on the food cup (feeding duration) and had significantly shorter latencies until climbing onto the cup than the singly reared pups.

Discussion

It is apparent from both the huddling and food competition tests that the presence of littermates throughout the preweaning period influences subsequent interactions with unfamiliar age-mates. In comparison to singly reared animals, those that had been raised in litters of 2 to 4 displayed

TABLE 3
Results of Feeding Competition Tests for Pairs Each Comprised of 1 Singly Reared and 1 Littermate-Reared Weanling^a

	Rearing condition	
	Pups reared without littermates	Pups reared with littermates
$\bar{X}$ Feeding duration (sec)	186.7 (137.2) ^b	292.5 (234.0)
	<i>*p</i> < .05	
$\bar{X}$ Latency until climbing onto food cup (sec)	700.3 (336.1)	492.9 (392.5)
	<i>*p</i> < .025	

^a *N* = 10 pairs.

^b Standard deviation in parentheses.

* One-tailed Wilcoxon matched-pairs signed-ranks test.

more huddling at 14–17 days of age and were more successful in a postweaning food competition task. The food competition data tend to corroborate those of Seitz (1954) who found that *adult* rats that had been reared in large litters were superior to rats from small litters in a somewhat similar competitive feeding test.

The observed differences between pups reared with vs without littermates are likely to have been mediated at least partially through the direct influence of siblings on the behavioral and physiological development of pups in the former condition. In addition to being continuously exposed to stimuli emanating from their littermates, animals growing up with agemate conspecifics engage in frequent overt interactions with their siblings—including huddling and agonistic or competitive interactions (e.g., Seitz, 1954). Moreover littermate stimulation may differ both quantitatively and qualitatively from stimulation provided by the mother (LaBarba & White, 1971). As suggested above, pups reared with littermates should therefore be more socially experienced and competent than singly reared pups and react differently than singletons with unfamiliar agemates.

Aside from the direct influence of littermates on one another's behavioral development, multiple-pup litters may elicit maternal behavior differing from that exhibited by mothers caring for single offspring. Subsequent differences in the social behavior of littermate vs singly reared pups could thereby result in turn from such differential maternal treatment interacting with direct littermate effects. Although mother–infant interactions for *A. cahirinus* females caring for one vs several offspring have not been investigated, litter size has been found to effect various aspects of maternal care in rats (Grota, 1973; Leigh & Hofer, 1973), mice (Priestnall, 1972), and gerbils (Elwood & Broom, 1978). It is further possible that the effects of litter size on offspring behavior in these same species are mediated somewhat by altered physiological development (e.g., rate of weight gain and development of the autonomic nervous system) resulting from differential maternal care (Leigh & Hofer, 1973). Pups growing up with littermates are exposed to an interacting constellation of sibling and maternal stimulation which differs considerably from the social stimulation provided solely by the mother of a single pup.

REFERENCES

- Alberts, J. R. (1978). Huddling by rat pups: Group behavioral mechanisms of temperature regulation and energy conservation. *Journal of Comparative and Physiological Psychology*, **92**, 231–245.
- Alberts, J. R., & Brunjes, P. C. (1978). Ontogeny of thermal and olfactory determinants of huddling in the rat. *Journal of Comparative and Physiological Psychology*, **92**, 897–906.
- Calhoun, J. B. (1962). *The ecology and sociology of the Norway rat*. Bethesda: Department of Health, Education and Welfare.

- Elwood, R. W., & Broom, D. M. (1978). The influence of litter size and parental behaviour on the development of Mongolian gerbil pups. *Animal Behaviour*, **26**, 438-454.
- Fleischer, S. F., & Turkewitz, G. (1979). Effect of neonatal stunting on development of rats: Large litter rearing. *Developmental Psychobiology*, **12**, 137-149.
- Galef, B. G., Jr. (1981). The ecology of weaning: Parasitism and the achievement of independence by altricial animals. In D. J. Gubernick & P. H. Klopfer (Eds.), *Parental Care in Mammals*, pp. 211-241. New York: Plenum.
- Grota, L. J. (1973). Effects of litter size, age of young, and parity on foster mother behaviour in *Rattus norvegicus*. *Animal Behaviour*, **21**, 78-82.
- Hennessy, M. B., Li, J., Lowe, E. L., & Levine, S. (1980). Maternal behavior, pup vocalizations, and pup temperature changes following handling in mice of 2 inbred strains. *Developmental Psychobiology*, **13**(6), 573-584.
- Hutchings, D. E. (1967). Infantile handling and body temperature change in rats: Effects of duration of cold exposure, rate of heat loss, level of hypothermia, and rewarming, on later emotionality. *Psychological Reports*, **21**, 985-1002.
- LaBarba, R. C., & White, J. L. (1971). Litter size variations and emotional reactivity in BALB/c mice. *Journal of Comparative and Physiological Psychology*, **75**, 254-257.
- Leigh, H., & Hofer, M. (1973). Behavioral and physiologic effects on littermate removal on the remaining single pup and mother during the pre-weaning period in rats. *Psychosomatic Medicine*, **35**, 497-508.
- Parker, H. B., Smith, R. F., Erickson, R. E., & Cromer, C. C. (1980). Clumping: A clue to the limitations of rodent density studies and their interpretation. *Behavioral and Neural Biology*, **28**, 335-342.
- Porter, R. H., Cavallaro, S. A., & Moore, J. D. (1980). Developmental parameters of mother-offspring interactions in *Acomys cahirinus*. *Zeitschrift fur Tierpsychologie*, **53**, 153-170.
- Porter, R. H., & Wyrick, M. (1979). Sibling recognition in spiny mice (*Acomys cahirinus*): Influence of age and isolation. *Animal Behaviour*, **27**, 761-766.
- Porter, R. H., Wyrick, M., & Pankey, J. (1978). Sibling recognition in spiny mice (*Acomys cahirinus*). *Behavioral Ecology and Sociobiology*, **3**, 61-68.
- Priestnall, R. (1972). Effects of litter size on the behaviour of lactating female mice (*Mus musculus*). *Animal Behaviour*, **20**, 386-394.
- Seitz, P. F. D. (1954). The effects of infantile experiences upon adult behavior in animal subjects: I. Effects of litter size during infancy upon adult behavior in the rat. *American Journal of Psychiatry*, **110**, 916-928.
- Swanson, L. J., & Campbell, C. S. (1980). Weaning in the female hamster: Effect of pup age and days postpartum. *Behavioral and Neural Biology*, **28**, 172-182.
- Trivers, R. L. (1974). Parent-offspring conflict. *American Zoologist*, **14**, 249-264.